# MTH 203: Sequences and series

*Mid-examination 1: 03/09/2025, Time: 12:00 to 13:00*

*Total points: 25*

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**Problem 1.** (2 points each)

(a) State the Axiom of Completeness.
(b) What is the Archimedean Property of the real numbers?

*Solution.*
*Part (a)*: *Axiom of Completeness* If $S \subset \mathbb{R}$ is any non-empty subset that is bounded above, then $S$ has a least upper bound in $\mathbb{R}$.

*Part (b)*: *Archimedean property:* If $x$ is any element of $\mathbb{R}$, there exists $n \in \mathbb{N}$ such that $n > x$. $\square$

**Problem 2.** (4 points) Let $A = \{x \in \mathbb{R} | -1 \leqslant x \leqslant 1\}$. Let $R \subset A \times A$ be a relation from $A$ to itself defined as follows:

$$\{(x, y) | x^2 + y^2 = 1\}.$$

Is $R$ a function? You must justify your answer.

*Solution.* The relation $R$ is a function if and only if for every $x \in A$, there exists a unique $y \in A$ such that $(x, y) \in R$. However, this is not true.

Indeed, if we take $x = 0$, then we see that $0^2 + 1^2 = 1$ and $0^2 + (-1)^2 = 1$. Thus, both $(0, 1)$ and $(0, -1)$ are in $\mathbb{R}$. Thus, $R$ is not a function. $\square$

**Problem 3.** (5 points) Let $S \subset \mathbb{R}$ and let $s = \sup(S)$. Let $t$ be an element of $\mathbb{R}$ such that $t \notin S$. Prove that $\sup(S \cup \{t\}) = \sup\{s, t\}$.

*Solution.* Let $u = \sup\{s, t\}$. Then, $u \geqslant s$ and $u \geqslant t$. Since $s$ is the supremum of $S$, $s \geqslant x$ for any $x \in S$. Thus, for any $x \in S$, using the inequalities $u \geqslant s$ and $s \geqslant x$, we conclude that $u \geqslant x$ (using the *transitivity* property of order relations). Since $u$ is greater than every element of $S$ and since $u \geqslant t$, we conclude that $u$ is an upper bound for $S \cup \{t\}$.

We will now prove that $u$ is a least upper bound for $S \cup \{t\}$. Let $v$ be any upper bound of $S \cup \{t\}$. Then, $v \geqslant x$ for any $x \in S$. Thus, since $s$ is the least upper bound of $S$, we have $v \geqslant s$. Also, since $v$ is an upper bound of $S \cup \{t\}$, $v \geqslant t$. Thus, $v$ is greater than every element of the set $\{s, t\}$. Since $u$ is the least upper bound of $\{s, t\}$, it follows that $v \geqslant u$. This shows that $u$ is the least upper bound of $S \cup \{t\}$. $\square$

**Problem 4.** (6 points) Let $A$ and $B$ be subsets of $\mathbb{R}$. Let $a = \sup(A)$ and let $b = \sup(B)$. Let $C$ be the subset defined as follows:

$$C = \{z \in \mathbb{R} | z = x + y \text{ for some } x \in A \text{ and } y \in B\}$$

Prove that $C$ is bounded above and $\sup(C) = a + b$.

*Solution.* Let $z$ be any element of $C$. Then, there exist $x \in A$ and $y \in B$ such that $z = x + y$. Since $a$ is an upper bound for $A$, we have $a \geqslant x$. Similarly, since $b$ is an upper bound for $B$, we have $b \geqslant y$. Adding these two inequalities, we get $a + b \geqslant x + y = z$. Since $z$ was an arbitrary element of $C$, we see that $a + b$ is an upper bound for $C$. Thus, $C$ is bounded above.

Now we will show that $a + b$ is the least upper bound of $C$. It suffices to show that if $\epsilon > 0$, then there exists an element $z$ of $C$ such that $z > a + b - \epsilon$.

As $\epsilon > 0$, and $1/2 > 0$ we have $\epsilon/2 = \epsilon \times (1/2) > 0$. Thus, since $a$ is the least upper bound of $A$, there exists $x \in A$ such that $x > a - \epsilon/2$. Similarly, since $b$ is the least upper bound of $B$, there exists $y \in B$ such that $y > b - \epsilon/2$. Adding the inequalities

$$x > a - \epsilon/2 \qquad \text{and} \qquad Y > b - \epsilon/2,$$

we get

$$x + y \geqslant a + b - \epsilon/2 - \epsilon/2 = a + b - \epsilon.$$

By the definition of $C$, the element $x + y$ is in $C$. Thus, we have found the element $x + y$ of $C$ which is greater than $a + b - \epsilon$. This completes the proof. □

**Problem 5.** (6 points) Let $S = \{1/n + 1/m | m \in \mathbb{N} \text{ and } n \in \mathbb{N}\}$. Compute $\inf(S)$ and $\sup(S)$.

*Solution.* If $m \in \mathbb{N}$, then $m \geqslant 1$. Since $m > 0$, $1/m > 0$. Multiplying the inequality $m \geqslant 1$ by $1/m$, we get $m \times (1/m) \geqslant 1 \times (1/m)$. Thus, $1 \geqslant 1/m$.

Thus, if $m$ and $n$ are in $\mathbb{N}$, we have $1 \geqslant 1/m$ and $1 \geqslant 1/n$. Adding these inequalities, we get $2 \geqslant 1/m + 1/n$. Thus, 2 is an upper bound of the set $S$. Since $2 = 1/1 + 1/1$, we also have $2 \in S$. Thus, for any upper bound $u$ of $S$, we have $u \geqslant 2$. Thus, 2 is the least upper bound of $S$, i.e. $\sup(S) = 2$.

If $m$ and $n$ are in $\mathbb{N}$, we have $1/m > 0$ and $1/n > 0$. Adding these two inequalities, we see that $1/m + 1/n > 0$. Thus, 0 is a lower bound for $S$. We claim that this is also the greatest lower bound. For this, it suffices to prove that if $\epsilon > 0$, then there exist $m$ and $n$ in $\mathbb{N}$ such that $1/m + 1/n < \epsilon$.

For this, we will choose $m$ and $n$ such that $1/m < \epsilon/2$ and $1/n < \epsilon/2$. The inequality $1/m < \epsilon/2$ holds if and only if $m > 2/\epsilon$. By the Archimedean property, there exists a natural number $N$ such that $1/N > 2/\epsilon$. We take $m = n = N$. Thus,

$$1/m + 1/n = 1/N + 1/N < \epsilon/2 + \epsilon/2 = \epsilon.$$

Thus, we see that $0 = \inf(S)$.
*Remark:* You may also use Problem 4 to solve this problem. Let $T = \{1/m | m \in \mathbb{N}\}$. If you take both $A$ and $B$ equal to $T$ in the statement of problem 4, the corresponding set $C$ is equal to $S$. Thus, as $\sup(T) = 1$, we see that $\sup(S) = 1 + 1 = 2$.

The same argument works for the infimum, but then one should explicitly justify that the analogue of Problem 4 works for the infimum. You may do this by explicitly seeing that the argument can be suitably modified for the infimum, or by applying the formula $\inf(S) = -\sup(-S)$. $\square$